

Nifty 100 Financial Analytics Platform

Institutional Technical Manual, Quantitative Specification & Operational Guide

1. Executive Summary & Platform Overview

The Nifty 100 Financial Data Pipeline & Analytics Platform is an institutional-grade financial intelligence engine engineered for quantitative screening, ratio computation, sector benchmarking, machine learning clustering, automated PDF report generation, and high-performance RESTful API serving.

Encompassing all 92 Nifty 100 constituent companies across 11 broad sectors over a 10-year historical timeline (2015–2024), the system parses multi-sheet Excel statements; enforces 16 strict data-quality rules; computes 20+ financial KPIs; ranks peer groups across 10 percentile dimensions; segments companies into 5 financial archetypes via KMeans; and generates executive 2-page PDF tearsheets.

2. System Architecture & Component Mapping

The repository is organized modularly under src/ into 7 core layers with decoupled interfaces:

Layer	Module Location	Core Functions & Capabilities
ETL Engine	src/etl/	Excel extraction, year/ticker normalization, SQLite loading, 16 DQ rules.
Analytics	src/analytics/	Ratios, 5y CAGR, Cash Flow KPIs, Valuation, KMeans clustering.
Screener	src/screener/	6 preset filters, winsorized quality score, threshold engine.
Dashboard	src/dashboard/ & pages/	8-screen Streamlit application served on port 8501.
NLP Engine	src/nlp/	Text regex parsing, 24-rule pros/cons confidence generator.
Report Engine	src/reports/	ReportLab 2-page tearsheets, sector reports, portfolio summary PDF.
REST API	src/api/	16 FastAPI endpoints served on port 8000 with OpenAPI specs.

3. Data Ingestion & Data-Quality Validation Framework

The ETL engine ingests financial statements from Excel workbooks in data/raw/100/. Data is cleaned via src/etl/normaliser.py, validated via src/etl/validator.py, and loaded into db/nifty100.db.

3.1 Data Quality Rules (DQ-01 to DQ-16)

16 automated rules check integrity before data is committed. Violations are logged to output/validation_failures.csv:

Rule ID	Severity	Validation Check Target Description
DQ-01	CRITICAL	Primary Key uniqueness on companies table (id).
DQ-02	CRITICAL	Composite Primary Key uniqueness (company_id, year) across financial tables.
DQ-03	CRITICAL	Foreign Key referential integrity to companies table.
DQ-04	ERROR	Balance sheet equation check: Assets == Liabilities + Equity.
DQ-05	ERROR	Profit & Loss check: Net Profit == Revenue - Expenses - Tax.
DQ-06	WARNING	Cash Flow check: Ending Cash == Operating + Investing + Financing.
DQ-07	WARNING	Non-negative check for Revenue, Total Assets, Market Cap.
DQ-08	INFO	Outlier detection: 5yr Revenue CAGR > 100% or < -50%.

4. Ratio Engine & Financial KPI Calculation Standard

Financial ratios are computed in `src/analytics/ratios.py` and populated in the `financial_ratios` table. Key ratio calculation methodologies include:

- **Return on Equity (ROE)**: $\text{Net Profit} / \text{Average Shareholders Equity}$.
- **Return on Capital Employed (ROCE)**: $\text{EBIT} / \text{Capital Employed}$.
- **Debt-to-Equity (D/E)**: $\text{Total Borrowings} / \text{Net Worth}$.
- **Interest Coverage Ratio (ICR)**: $\text{EBIT} / \text{Interest Expense}$.
- **Asset Turnover**: $\text{Total Sales} / \text{Average Total Assets}$.
- **Free Cash Flow (FCF)**: $\text{Operating Cash Flow} - \text{Capital Expenditure}$.

5. Quality Screener & Composite Quality Scoring Engine

The screener engine (`src/screener/engine.py`) parses `config/screener_config.yaml` and applies threshold filters. The Winsorized Composite Quality Score (0–100) ranks companies based on relative performance across core financial dimensions.

6. Screener Preset Definitions & Threshold Specs

Preset Name	Core Filter Criteria & Threshold Rules
Quality Compounders	ROE ≥ 15%, Debt/Equity ≤ 1.0, 5y Rev CAGR ≥ 10%, Positive FCF.
High Dividend Yield	Dividend Yield ≥ 3.0%, Debt/Equity ≤ 1.5, Positive Net Profit.
Deleveraging Champions	Debt/Equity decreased YoY, Operating Cash Flow > 0.
Undervalued Growth	P/E ≤ Sector Median, 5y PAT CAGR ≥ 12%, ROE ≥ 12%.
Capital Allocators	CapEx Intensity ≥ 10%, ROE ≥ 14%, CFO Quality Score ≥ 0.8.
Cashflow Fortress	Free Cash Flow > 0, FCF Conversion Rate ≥ 70%, ICR ≥ 4.0.

7. Peer Group Percentile Ranking Methodology

In `src/analytics/peer.py`, companies are mapped to 11 peer groups. 10 financial metrics are converted into percentile ranks (0–100%). Metrics where lower is superior (such as Debt-to-Equity) are automatically inverted so that lower values receive higher ranks.

8. Machine Learning Clustering & Financial Archetypes

In `src/analytics/clustering.py`, KMeans (`k=5`, `random_state=42`) clusters companies using 5 standardized features (ROE, Debt/Equity, 5y Rev CAGR, 5y PAT CAGR, OPM). The output is written to `output/cluster_labels.csv`:

Cluster ID	Archetype Name	Financial Profile Characteristics
Cluster 0	High-Quality Compounders	High ROE (>20%), Low D/E (<0.5), Strong FCF & OPM.
Cluster 1	Emerging Growth	High Revenue CAGR (>15%), Moderate ROE, Active CapEx.
Cluster 2	Defensive Dividend Payers	Stable Cash Flow, High Dividend Yield, Low Beta.
Cluster 3	Value Cyclicals	Low P/E, Cyclical Earnings, Moderate Leverage.
Cluster 4	Distressed or Turnaround	Low ROE (<8%), High D/E (>1.5), Negative FCF.

9. Streamlit Multi-Page Web Application Architecture

The Streamlit dashboard (streamlit run src/dashboard/app.py) provides 8 interactive analytics screens on port 8501:

1. **Home Screen (01_home.py)**: 6 top KPI tiles, sector breakdown donut chart, top compounders.
2. **Company Profile (02_profile.py)**: Corporate card, 10-year P&L, ROE/ROCE charts, NLP pros/cons badges.
3. **Financial Screener (03_screener.py)**: 10 dynamic sliders, 6 preset triggers, live count, CSV download.
4. **Peer Comparison (04_peers.py)**: 8-metric Plotly radar chart vs peer group average and benchmark.
5. **Financial Trends (05_trends.py)**: 10-year multi-metric YoY overlay charts.
6. **Sector Analytics (06_sectors.py)**: 4D sector bubble chart and median KPI bar charts.
7. **Capital Allocation (07_capital.py)**: Treemap of companies grouped by capital allocation archetypes.
8. **Annual Reports (08_reports.py)**: BSE annual report link repository and PDF tearsheet downloads.

10. FastAPI REST API Specification (16 Endpoints)

The REST API is served via FastAPI on <http://localhost:8000/api/v1>. OpenAPI documentation is at </docs>.

HTTP Method	Endpoint Path	Description & Response Specs
GET	/api/v1/health	System health status, database row counts, uptime.
GET	/api/v1/companies	List all 92 companies with sector & search filters.
GET	/api/v1/companies/{ticker}	Full company profile and latest financial ratios.
GET	/api/v1/companies/{ticker}/pl	Historical Profit & Loss statements.
GET	/api/v1/companies/{ticker}/bs	Historical Balance Sheet statements.
GET	/api/v1/companies/{ticker}/cashflow	Historical Cash Flow statements.
GET	/api/v1/companies/{ticker}/ratios	Computed ratio history per year.
GET	/api/v1/companies/{ticker}/tearsheet	Binary download of 2-page PDF tearsheet.
GET	/api/v1/screener	Execute multi-parameter quality screener.
GET	/api/v1/sectors	Summary and median KPIs across 11 broad sectors.
GET	/api/v1/sectors/{sector}/companies	All constituent companies in a sector.
GET	/api/v1/peers/{group_name}	Peer group members and percentile ranks.
GET	/api/v1/companies/{ticker}/peers/compare	Radar chart comparative metrics vs peer avg.
GET	/api/v1/market-cap/{ticker}	Historical valuation multiples (P/E, P/B, EV/EBITDA).
GET	/api/v1/portfolio/stats	Portfolio P10–P90 percentile statistics table.
GET	/api/v1/companies/{ticker}/documents	Annual report BSE document URLs.

11. API cURL Examples & Code Snippets

Sample cURL commands for REST endpoint integration:

1. Health Endpoint

```
curl -X GET 'http://localhost:8000/api/v1/health'
```

2. Quality Screener ($ROE \geq 15\%$, $D/E \leq 1.0$)

```
curl -X GET 'http://localhost:8000/api/v1/screener?min_roe=15.0&max_de=1.0'
```

3. Fetch TCS Company Profile

```
curl -X GET 'http://localhost:8000/api/v1/companies/TCS'
```

4. Download TCS Tearsheet PDF

```
curl -X GET 'http://localhost:8000/api/v1/companies/TCS/tearsheet' --output TCS_tearsheet.pdf
```

12. ReportLab PDF Tearsheet Generation System

The PDF generation suite (src/reports/tearsheet.py, batch_generator.py) builds institutional 2-page company tearsheets (>100KB each) featuring embedded Matplotlib vector/PNG charts (P&L; bar chart, ROE/ROCE line chart, Balance Sheet stacked bar chart, and Cash Flow waterfall chart).

13. Operational Guide, Testing & Troubleshooting

13.1 Running the Automated Test Suite (250 Tests)

Run the full unit and API test suite to generate the HTML test report:
`PYTHONPATH=. venv/bin/pytest tests/ --html=reports/pytest_report.html -v`

13.2 Launching Services Simultaneously

- Launch Streamlit Dashboard (Port 8501):
`venv/bin/streamlit run src/dashboard/app.py`
- Launch FastAPI REST Server (Port 8000):
`venv/bin/uvicorn src.api.main:app --port 8000 --reload`

13.3 Troubleshooting Matrix

Symptom / Issue	Likely Cause	Resolution Action
HTTP 404 on API endpoint	Invalid ticker symbol	Verify ticker against /api/v1/companies.
Port 8000 already in use	Another uvicorn instance	Run <code>lsof -i :8000</code> and kill PID.
Missing PDF tearsheets	Batch generator not run	Execute <code>python src/reports/batch_generator.py</code> .
SQLite database locked	Concurrent write transaction	Ensure read-only connections are closed.

14. Portfolio Percentile Statistics Summary (P10–P90)

Statistical summary across all 92 Nifty 100 constituent companies (written to output/portfolio_stats.csv):

KPI Metric	Mean	Std	P10	P25	P50 (Median)	P75	P90
Return on Equity (%)	16.8%	10.2%	5.1%	10.2%	15.4%	22.1%	31.2%
ROCE (%)	18.4%	11.5%	6.2%	11.5%	17.1%	24.5%	34.1%
Net Profit Margin (%)	13.5%	8.7%	3.1%	7.5%	12.2%	18.4%	26.0%
Operating Margin (%)	21.2%	12.4%	8.5%	13.2%	19.5%	27.8%	38.5%
Debt-to-Equity	0.65	0.82	0.00	0.05	0.32	0.85	1.80
5yr Revenue CAGR (%)	11.2%	6.8%	3.5%	7.1%	10.8%	15.2%	20.5%
5yr PAT CAGR (%)	13.8%	11.4%	1.2%	6.8%	12.5%	19.2%	28.4%

15. Outlier Detection Engine Specs

Outliers are flagged per broad sector when a company's KPI metric z-score $|Z| > 3.0$. Results are exported to output/outlier_report.csv for quantitative risk monitoring.

16. Final Project Acceptance Gates Verification (AC-01 to AC-20)

All 20 formal acceptance criteria have been verified and signed off for Day 45 final project delivery:

Gate ID	Acceptance Criteria Description	Verification Status
AC-01	SELECT COUNT(*) FROM companies == 92	PASS ■
AC-02	≥ 90% companies have ≥ 10 years P&L, BS, CF records	PASS ■
AC-03	PRAGMA foreign_key_check == 0 rows	PASS ■
AC-04	SELECT COUNT(*) FROM financial_ratios ≥ 1100	PASS ■
AC-05	Revenue CAGR spot-check within 0.1% of manual Excel	PASS ■
AC-06	ROE matches companies.roe_percentage within 5% for 5 companies	PASS ■
AC-07	Quality screener returns 10–50 companies	PASS ■
AC-08	Company Profile screen loads < 3 seconds	PASS ■
AC-09	CSV download from screener is valid and well-formed	PASS ■
AC-10	No text overflow in sampled tearsheet PDFs	PASS ■
AC-11	GET /api/v1/health returns HTTP 200	PASS ■
AC-12	TCS ratios endpoint returns 10+ years data	PASS ■
AC-13	API screener results match screener_output.xlsx	PASS ■
AC-14	peer_percentiles table has data for all 11 peer groups	PASS ■
AC-15	All 92 companies have cluster_id in cluster_labels.csv	PASS ■
AC-16	All 92 companies have 1+ pro and 1+ con in pros_cons_generated.csv	PASS ■
AC-17	92 tearsheet PDFs exist in reports/tearsheets/, each ≥ 30KB	PASS ■
AC-18	pytest shows 60+ tests collected, 0 failures	PASS ■ (250 Passed)
AC-19	validation_failures.csv has required schema	PASS ■
AC-20	analyst_guide.pdf is 10+ pages	PASS ■

17. Final Sign-off & Release Summary

The Nifty 100 Financial Data Pipeline Capstone Project is officially complete and signed off as of Day 45. All 23 deliverables are archived in output/final_deliverables/ and tagged release v6.0.